

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388318
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Bernard; Clement et al.

Cover for a rotating electric machine

Abstract

A cover for a rotating electric machine includes an axis of rotation (X), the cover being designed to be arranged on an inverter of the machine. The cover includes a skirt extending in a radial direction and forming a perimeter of the cover, the perimeter defining an inner portion of the cover and an open area arranged in the inner portion and comprising at least one axial opening. The open area extends over at least 50% of the inner portion of the cover.

Inventors:	Bernard; Clement (Creteil, FR), Walme; Benoit (Creteil, FR), Zanella; Charlie (St Quentin Fallavier, FR), Beaucote; Sebastien (St Quentin Fallavier, FR), Dumontagne; David (Creteil, FR), Dumez; Charles (Creteil, FR), Da Cruz Rebelo; Mario (Creteil, FR), Azizi Samir; Abdelaziz (Creteil, FR), Falguier; Manuel (Creteil, FR)
Applicant:	VALEO EQUIPEMENTS ELECTRIQUES MOTEUR (Creteil, FR)
Family ID:	70614316
Assignee:	VALEO EQUIPEMENTS ELECTRIQUES MOTEUR (Creteil, FR)
Appl. No.:	17/907641
Filed (or PCT Filed):	March 09, 2021
PCT No.:	PCT/EP2021/055857
PCT Pub. No.:	WO2021/197770
PCT Pub. Date:	October 07, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230155447 A1	May. 18, 2023

Foreign Application Priority Data

FR

2003283

Apr. 02, 2020

Publication Classification

Int. Cl.: H02K5/20 (20060101); H02K9/04 (20060101)

U.S. Cl.:

CPC H02K5/207 (20210101); H02K9/04 (20130101); H02K2213/03 (20130101)

Field of Classification Search

CPC: H02K (5/207); H02K (5/18); H02K (5/20); H02K (5/22); H02K (5/225); H02K (9/04); H02K (9/06); H02K (11/30); H02K (11/33); H02K (2213/03)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0084197	12/2012	Mueller	417/366	F04D 29/5813
2013/0257193	12/2012	Toda et al.	N/A	N/A
2019/0190351	12/2018	Gregorich et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3 503 355	12/2018	EP	N/A
2950752	12/2010	FR	H02K 11/048
2012/143035	12/2011	JP	N/A
2018/084152	12/2017	JP	N/A
2018207640	12/2017	JP	N/A
WO-2018138448	12/2017	WO	H02K 11/048

OTHER PUBLICATIONS

Machine Translation of JP_2018207640_A (Year: 2018). cited by examiner

Machine Translation of WO_2018138448_A1 (Year: 2018). cited by examiner

Machine Translation of FR_2950752_A1 (Year: 2011). cited by examiner

International Search Report issued May 26, 2021 in PCT/EP2021/055857 filed on Mar. 9, 2021, 2 pages. cited by applicant

Office Action issued in corresponding Japanese Patent Application No. 2022-559982 on Nov. 7, 2023 (English translation only); pp. 1-5. cited by applicant

Primary Examiner: Singh; Alexander A

Assistant Examiner: Mann, Jr.; Charlie Frank

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

- (1) The invention relates, in particular, to a cover for a rotating electric machine.
- (2) The invention is particularly advantageously applicable in the field of rotating electric machines such as alternators, starter-alternators, or even reversible machines or electric motors. By way of a reminder, a reversible machine is a rotating electric machine capable of working in a reversible manner, on the one hand, as an electric generator in the alternator function and, on the other hand, as an electric motor, for example, in order to start the heat engine of the vehicle such as a motor vehicle.

Description of the Related Art

- (3) A rotating electric machine comprises a rotor that is free to rotate about an axis, and a fixed stator. In alternator mode, when the rotor is rotating, it induces a magnetic field at the stator, which converts it into electric current in order to supply power to the electrical consumers of the vehicle and to recharge the battery. In motor mode, the stator is electrically powered and induces a magnetic field that rotates the rotor, for example, in order to start the heat engine.
- (4) The assembly formed by the rotor and the stator is arranged in a casing on which an inverter is arranged. The inverter, which forms an electronic assembly of the machine, is conventionally protected by a cover. The cover has a plate extending in a radial direction to cover an axial end surface of the inverter and a skirt extending in an axial direction from the plate to cover a circumferential periphery of the inverter. The plate and/or the skirt may have openings allowing a cooling air flow to pass through.
- (5) Such a cover does not have an optimized shape helping to protect the elements of the rotating electric machine without degrading performances in terms of cooling said elements.

BRIEF SUMMARY OF THE INVENTION

- (6) The aim of the present invention is to help avoid the drawbacks of the prior art by proposing an optimized cover shape.
- (7) To this end, the present invention therefore relates to a cover for a rotating electric machine comprising an axis of rotation. The cover is intended to be arranged on an inverter of said machine and comprises a skirt extending in an axial direction and forming a perimeter of the cover, the perimeter defining an inner portion of the cover and an open region arranged in the inner portion and comprising at least one axial opening. According to the present invention, the open region extends over at least 50% of the inner portion of the cover.
- (8) The presence of such an open region on the cover helps considerably improve the passage of the cooling air flow, thus allowing better cooling of the inverter without affecting the mechanical strength of the cover or the protection of the elements of the machine against the external environment. Indeed, enlarging the area of the opening in the cover makes it possible, inter alia, to facilitate the circulation of the air flow between the cover and the inverter and, in particular, on the axial end surface of the inverter. Moreover, this novel cover shape also helps to reduce the axial space requirement of the machine, as well as its weight, and to simplify the arrangement of the inverter and in particular of the connector for connecting the inverter to the general computer of the

vehicle in which the machine is mounted.

(9) According to one embodiment, the open region extends over at least 70% of the inner portion of the cover and in particular 84%. This increase in the dimensions of the open region helps further optimize the shape of the cover in order to improve the cooling of the inverter.

(10) According to one embodiment, the open region extends over the whole of the area of the inner portion.

(11) According to one embodiment, the axial opening of the open region extends over at least 50% of the inner portion of the cover. It should be understood here that the cover has at least one opening extending over at least 50% of the inner portion of the cover. The cover may also have other openings.

(12) According to one embodiment, the open region comprises several axial openings. This makes it possible to enlarge the open region while maintaining the possibility of adding elements such as protection, gripping or mechanical strength elements in the inner portion. Alternatively, the open region may be formed from a single axial opening.

(13) According to one embodiment, the cover comprises at least one axial protection portion intended to be arranged axially opposite at least one phase connection of the machine, the axial protection portion extending into the inner portion. The axial protection portion helps protect the electrical connection between the phases of the stator and the inverter from the external environment in order to ensure safe operation of the machine.

(14) For example, the axial protection portion extends from the skirt.

(15) According to one embodiment, the skirt comprises at least one radial protection portion intended to be arranged radially opposite at least one phase connection of the machine. This radial protection portion helps protect the phase connection of the machine from the external environment.

(16) According to one embodiment, the skirt comprises at least two radial protection portions, each being intended to be arranged radially opposite at least one phase connection of the machine, and at least one connecting portion extending circumferentially between said two radial protection portions, the connecting portion having a height, in an axial direction, less than the height, in an axial direction, of the radial protection portions. The difference in height between the radial protection portions and the connecting portion makes it possible to arrange a primary radial opening region in the cover. The radial opening helps improve the cooling of the inverter. For example, the primary radial opening region is formed from one or more primary radial openings.

(17) According to one embodiment, the height of the connecting portion is less than half the height of a radial protection portion. In other words, the height of the primary radial opening region, in an axial direction, is greater than half the axial thickness of the radial protection portion.

(18) According to one embodiment, the skirt is formed from a succession of radial protection portions and connecting portions.

(19) According to one embodiment, each radial protection portion comprises a secondary radial opening region. This also helps improve the cooling of the machine. For example, the secondary radial opening region is formed from one or more secondary radial openings.

(20) For example, the area of the secondary radial opening region is between 15% and 25% of the area of the associated protection portion. This helps improve the cooling of the machine while ensuring the mechanical strength of the cover and protecting the machine from the external environment.

(21) According to one embodiment, the skirt forms a closed perimeter. This helps improve the mechanical strength of the cover.

(22) According to one embodiment, the cover further comprises at least one fastening portion, the fastening portion being intended to cooperate with a fastening element in order to mount the cover on an inverter of the rotating electric machine, the fastening portion extending into the inner portion.

- (23) For example, the axial fastening portion extends from the skirt or from the axial protection portion or from the gripping means.
- (24) According to one embodiment, the fastening portion extends in a radial direction and comprises an opening through which the fastening element can pass.
- (25) According to one embodiment, the cover further comprises at least one gripping means extending into the inner portion. This gripping means makes it possible, during the process of assembling the machine, to simplify the arrangement of the cover on the inverter before they are fastened together.
- (26) For example, the gripping means extends from the skirt or from the fastening portion or from the axial protection portion.
- (27) According to one embodiment, the gripping means is formed from a planar portion extending in a radial plane. This makes it possible to move the cover without increasing the axial space requirement of the machine. This may also provide a labelling area for recording the references of the machine.
- (28) According to one embodiment, a set-back portion is formed at at least one end of the gripping means so that said means extends axially at a distance from the skirt. This helps reduce the axial height of the skirt in order to improve the cooling of the machine.
- (29) The present invention also relates to a rotating electric machine comprising an inverter and a cover as previously described, the cover being mounted on the inverter. The rotating electric machine can, advantageously, form an alternator, a starter-alternator, a reversible machine or an electric motor.
- (30) According to one embodiment, the inverter comprises a power stage for supplying power to the phases of a stator of the machine and/or for rectifying the current leaving said phases. The inverter further comprises a control stage for controlling the machine.
- (31) According to one embodiment, the inverter comprises at least one heat sink plate, said plate forming, at least partially, one axial end of said inverter and being arranged at least partially facing the open region. The heat sink plate helps to protect the elements of the inverter from the external environment and also to improve the cooling of the inverter by conduction.
- (32) According to one embodiment, the inverter comprises a capacitor bank and the open region is arranged so as to be at least partially axially aligned with the capacitor bank.
- (33) According to one embodiment, the machine comprises a casing arranged to at least partially surround a stator and a rotor of said machine, the inverter being arranged axially between the cover and the casing.
- (34) According to one embodiment, the rotating electric machine further comprises a shutter arranged at one axial end of the inverter, the shutter comprising a radially extending portion arranged opposite a capacitor bank of the inverter. The radially extending portion helps to protect the capacitor bank from the external environment.
- (35) According to one embodiment, the machine comprises a casing arranged to at least partially surround a stator and a rotor of said machine, the shutter being arranged axially between the casing and the inverter. The shutter makes it possible to at least partially cover axial openings arranged in the casing. The shutter therefore helps prevent a cooling air flow that has already been heated by the rotor or the stator from circulating in the inverter.
- (36) According to one embodiment, the power stage and the control stage are axially superposed so as to define a cooling passage between them, the primary radial opening region being in particular at least partially arranged facing said cooling passage.
- (37) According to one embodiment, the machine comprises a stator comprising at least one phase terminal and the inverter comprises at least one power module comprising at least one power terminal, the power terminal being electrically connected to the phase terminal in order to form a phase connection.
-

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The present invention may be better understood upon reading the following detailed description of nonlimiting exemplary embodiments of the invention and upon studying the appended drawings.
- (2) FIG. 1 shows a schematic and partial sectional view of a rotating electric machine according to one example of the invention.
- (3) FIG. 2 shows a schematic and partial perspective view of an example of a cover according to the invention mounted on an inverter.
- (4) FIG. 3 shows a schematic and partial perspective view of the cover shown in FIG. 2.
- (5) FIG. 4 shows a schematic and partial perspective view of an example of a shutter according to the invention.
- (6) FIG. 5 shows a schematic and partial perspective view of another example of a cover according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (7) Identical, similar or comparable elements use the same reference signs from one figure to the next. It also should be noted that the various figures are not necessarily to the same scale. In addition, the embodiments which are described hereinafter are in no way limiting. In particular, it is possible to conceive of variants of the invention comprising only a selection of features described hereinafter, in isolation from the other features described.
- (8) FIG. 1 depicts an example of a compact and polyphase rotating electric machine **10**, in particular for a vehicle such as a motor vehicle or a drone. This machine **10** converts mechanical energy into electrical energy, in alternator mode, and can operate in motor mode in order to convert electrical energy into mechanical energy. This rotating electric machine **10** is, for example, an alternator, a starter-alternator, a reversible machine or an electric motor.
- (9) In this example, the machine **10** comprises a casing **11** on which an inverter **36** is mounted. Inside this casing **11**, it further has a shaft **13**, a rotor **12** which rotates as one with the shaft **13** and a stator **15** surrounding the rotor **12**. The rotational movement of the rotor **12** takes place about an axis X. In the rest of the description, the axial direction corresponds to the axis X, which passes through the shaft **13** at the center thereof, while the radial orientations correspond to planes that are concurrent with, and in particular perpendicular to, the axis X. For the radial directions, the term inner corresponds to an element oriented toward the axis, or closer to the axis than a second element, and the term outer denotes distance from the axis.
- (10) In this example, the casing **11** comprises a front end plate **16** and a rear end plate **17** which are joined together. These end plates **16**, **17** are hollow in form and each centrally bear a bearing carrier coupled to a respective ball bearing **18**, **19** for the rotational mounting of the shaft **13**. Moreover, the casing **11** has fastening means (not shown) for mounting the rotating electric machine **10** in the vehicle.
- (11) A drive member such as a pulley **20** may be fastened to a front end of the shaft **13**. This member makes it possible to transmit the rotational movement to the shaft or allows the shaft to transmit its rotational movement to the belt. In the rest of the description, the terms front/rear refer to this member. Thus, a front face is a face oriented in the direction of the member, while a rear face is a face oriented in the direction away from said member.
- (12) The rear end of the shaft **13** bears, in this case, slip rings **21** belonging to a commutator **22**. A brush holder (not shown) comprises brushes arranged in such a way as to rub on the slip rings **21**. The brush holder is connected to the inverter **36** and, in particular, to a control module **23** of the inverter.
- (13) The front end plate **16** and the rear end plate **17** may have substantially lateral openings for the passage of an air flow in order to allow the machine **10** to be cooled by air circulation brought

about by the rotation of a front fan **25** arranged on a front axial face of the rotor **12** and of a rear fan **26** arranged on a rear axial face of said rotor.

(14) In this example, the rotor **12** is a claw-pole rotor comprising two pole wheels **31**. Each pole wheel **31** is formed from a plate oriented transversely, a plurality of claws forming magnetic poles, and a cylindrical core. The rotor has a coil **35** wound around the core. For example, the slip rings **21** belonging to the commutator **22** are connected by wire connections to said coil **35**. The rotor **12** may also comprise magnetic elements **32**, such as permanent magnets, interposed between two adjacent claws. Alternatively, the rotor may be formed from a stack of laminations housing permanent magnets forming the magnetic poles.

(15) In this exemplary embodiment, the stator **15** has a body **27** formed from a stack of laminations provided with slots, equipped with a slot insulator for the mounting of an electric winding **28**. The winding passes through the slots in the body **27** and forms a front overhang and a rear overhang on either side of the body of the stator. Furthermore, the winding **28** is formed from one or more phases comprising at least one electrical conductor. Each phase comprises an end forming a phase output **33** which is electrically connected to the inverter **36**.

(16) The inverter **36**, which is in this instance mounted on the casing **11**, forms the electronic assembly of the machine. The inverter comprises a power stage used to receive or supply an electric power signal from or to the electrical phases of the winding **28**. The power stage comprises at least one power module **24** comprising a power terminal **37** arranged to be electrically connected to a phase output **33** in order to form a phase connection **38**. The power module forms a bridge voltage rectifier for converting the AC voltage generated by the phases of the stator into a DC voltage and/or, conversely, for converting a DC voltage into an AC voltage to be fed to the phases of the stator. The inverter also comprises a control stage comprising a control module **23** making it possible, in particular, to regulate the voltage injected via the brush holder to rotor **12** and providing the interface with a computer external to the vehicle.

(17) In the example shown in FIGS. **1** and **2**, the power stage and the control stage are stacked on each other in an axial direction. More particularly in this example, the power stage comprises a heat sink, referred to as the first heat sink **39**, mounted on the casing **11**, the power module **24** being mounted on said first heat sink. Moreover, the control stage comprises a heat sink, referred to as the second heat sink **40**, mounted on the first heat sink **39**, the control module **23** being mounted on said second heat sink. For example, the two heat sinks are each formed from a thermally conductive material such as metal.

(18) The inverter **36** also comprises at least one capacitor (not shown) for filtering electrical signals from the inverter. The capacitor is arranged in a capacitor bank **34** which may contain one or more capacitors. The capacitor bank is arranged, for example, in a recess in the first heat sink **39**. Said recess may be closed by the second heat sink **40**. The capacitor bank **34** is, in this instance, arranged between the two heat sinks in order to improve the cooling of said bank.

(19) The machine **10** further comprises a cover **42** mounted on the inverter **36**. For example, the cover is formed from an electrically insulating material such as plastic. By way of further example, the cover is a one-piece component and is, in particular, formed as a single part, for example by molding.

(20) The cover **42** comprises a skirt **43** extending in an axial direction and forming a perimeter of the cover. In other words, the perimeter forms an outer periphery inside which an inner portion of the cover is defined. The cover **42** has an open region **44** arranged in the inner portion. The open region **44** extends over at least 50%, and in particular at least 70%, of an axial end area of the inner portion of the cover **42**. The axial end area of the inner portion of the cover is defined as being the area delimited by the perimeter of the skirt. In other words, the cover **42** is closed axially over an area covering less than half the total axial end area of said cover.

(21) In the example shown in FIG. **2**, the open region **44** extends over an area equal to 84% of the axial end area of the inner portion of the cover **42**. The temperature of the inverter **36** at a given

speed is, on average, with this example of the opening percentage of the open region, approximately 143.2° C., whereas the temperature of an inverter of the prior art comprising a conventional cover that has an opening percentage of less than 20% is approximately 147° C. With this exemplary embodiment of the invention, the temperature of the inverter **36** at a given speed has therefore dropped by 3.8° C. compared to a cover of the prior art.

(22) The open region **44** may be arranged in such a way as to be at least partially axially aligned with the capacitor bank **34**. Moreover, in this example, the inverter **36** comprises a heat sink plate **41** arranged to cover the control stage. For example, the heat sink plate **41** is formed from a metal material. More specifically, a thermal adhesive may be arranged between the part of the second heat sink **40** in contact with the capacitor bank **34** and the heat sink plate **41**. This helps improve the cooling of the capacitor bank by conduction. The fact that the cover **42** is axially open over at least 50% of its area and in particular above the capacitor bank **34** helps further improve the cooling of said bank. Indeed, the rear axial end surface of the heat sink plate **41** is exposed to a greater quantity of cooling air flow, and this also helps prevent hot air from stagnating axially between the cover **42** and said plate **41**.

(23) The open region **44** is arranged facing the heat sink plate **41**. Moreover, in this instance, a signal connector **46** is arranged to pass through said plate **41** and the open region **44** in order to connect the control module **23** to a computer external to the vehicle for controlling the machine.

(24) The open region **44** is formed from at least one axial opening **45**. In the example shown in FIGS. 2 and 3, the open region comprises three axial openings **45**. The openings **45** are in this instance of different shapes and sizes.

(25) The cover **42** may comprise at least one axial protection portion **47** arranged axially opposite at least one phase connection **38**. The axial protection portion extends into the inner portion from the skirt **43**. Said portion **47** extends in a radial direction. In the example shown in FIG. 2, the stator **15** comprises six phase outputs **33** grouped into pairs along the circumference of the machine. Therefore, in this instance, the cover **42** comprises three axial protection portions **47**, each being arranged axially opposite two phase connections **38**. Alternatively, each axial protection portion could be arranged opposite a single phase connection or a number of phase connections greater than two.

(26) The skirt **43** extends circumferentially so as to at least partially surround the inverter **36** and, in particular, the control stage. The skirt **43** comprises at least two radial protection portions **48** and at least one connecting portion **49**. Each radial protection portion **48** is arranged radially opposite at least one phase connection **38** of the machine. Preferably, each axial protection portion **47** extends from a radial protection portion **48**. In the example shown in FIG. 2, the skirt **43** comprises three radial protection portions **48**, each being arranged radially opposite two phase connections **38**. Alternatively, each radial protection portion could be arranged opposite a single phase connection or a number of phase connections greater than two.

(27) The connecting portion **39** extends circumferentially between two radial protection portions **48**. Therefore, the skirt is formed from a succession of radial protection portions **48** and connecting portions **49**. The connecting portions **49** extend, in particular, in such a way as to be aligned axially in relation to each other. For example, in this instance, the skirt **43** forms a closed annular perimeter.

(28) As shown in FIG. 3, the connecting portion **49** has, in this instance, a height **H2** less than the respective height **H1** of the radial protection portions **48**, the heights **H1**, **H2** being measured in an axial direction. An axial end of the connecting portion **49** is, preferably, arranged so as to be aligned in a radial plane with an axial end of the adjacent radial protection portions **48**.

(29) In the example shown in FIG. 2, the height **H2** of the connecting portion **49** is less than half the height **H1** of the radial protection portions **48**.

(30) The difference in height between the radial protection portions **48** and the connecting portion **49** makes it possible to form a primary radial opening region in the cover **42**. For example, the

primary radial opening region is formed from one or more primary radial openings **50**. Two radial protection portions **48** are separated from each other both by a connecting portion **49** and by a primary radial opening **50**. The stack of the power stage and the control stage defines a cooling passage **52** between said stages, which can be seen in FIG. 2. Each primary radial opening **50** is arranged at least partially facing said cooling passage **52**.

(31) In the example of FIGS. 2 and 3, each radial protection portion **48** may comprise a secondary radial opening region, for example formed from one or more secondary radial openings **51**. The secondary radial openings help improve the cooling of the inverter **36** and the phase connections **38** while allowing said connections to be protected from the external environment of the machine. For example, the area of the secondary radial opening region is between 15% and 25% of the area of the associated protection portion **48**.

(32) More specifically, in the example shown here, each secondary radial opening **51** extends from an axial end of the skirt and, in particular, from the associated axial protection portion **47**. With continuing reference to this example, the secondary radial opening region is formed from three secondary radial openings **51**. Two of said openings **51** are, in this instance, arranged axially between a respective phase connection **38** and the axial protection portion **47**. The other opening **51** is, in this instance, arranged radially between two phase connections **38**.

(33) The cover **42** may comprise several fastening portions **53** arranged to mount said cover on the inverter **36**. For this purpose, each fastening portion has, for example, an opening through which a fastening element **54** is inserted. The cover and the inverter are assembled together, in particular, by screwing or clip-fastening. In the example shown in FIG. 3, each fastening portion **53** extends radially from the skirt **43** into the inner portion of the cover **42** and, in particular, from a connecting portion **49**. Each fastening portion **53** may be partially surrounded by a rim **56** extending axially in particular to stiffen said fastening portion.

(34) The cover **42** may comprise a gripping means **55** making it possible, during the process of assembling the machine, to simplify the arrangement of the cover on the inverter **36** before they are fastened together. In the example shown in FIG. 3, the gripping means **55** is formed from a planar portion extending in a radial plane into the inner portion of the cover **42**. This makes it possible, in particular, to use an automatic gripping device to position the cover **42** on the inverter **36**. More specifically in this example, the cover comprises two gripping means **55**. The first gripping means extends between two fastening portions **53** and the second gripping means extends between a fastening portion **53** and an axial protection portion **47**. For example, each gripping means **55** extends axially at a non-zero axial distance from the skirt **43**. In other words, in this instance, each gripping means **55** extends in a radial plane not comprising the skirt **43**. For this purpose, each gripping means **55** extends more particularly from an axial end of the rim **56** partially surrounding the associated fastening portion **53** or from an axial set-back portion **57** formed between the axial protection portion **47** and said means **55**.

(35) As shown in FIG. 2, the axial end of the machine **10** is formed locally by several elements arranged in different radial planes: the signal connector **46**, the heat sink plate **41**, the cover **42** and, in particular, the gripping means **55** of the cover.

(36) As shown in FIGS. 1 and 4, the rotating electric machine **10** further comprises a shutter **58** arranged axially between the inverter **36** and the casing **11**. More specifically in this instance, the shutter is mounted sandwiched between the rear end plate **17** and the first heat sink **39** and may comprise openings for the passage of an element for fastening the inverter **36** with the casing **11**. The shutter may, moreover, comprise at least one damper **60** in order to help assemble the shutter in a more solid manner.

(37) For example, the shutter **58** has an annular shape and is arranged in such a way as to at least partially cover axial openings formed in the rear end plate **17**. The shutter **58** comprises openings **59** through which the phase outputs **33** of the stator can pass.

(38) In the example shown in FIG. 4, the shutter **58** comprises a radially extending portion **61**

arranged so as to be aligned in an axial direction with the capacitor bank 34 of the inverter 36. The radially extending portion 61 extends, in particular, in a radial direction, in such a way as to form an external radial projection of the shutter. For example, the radially extending portion 61 is surrounded on its outer periphery by a rim 62 extending axially.

(39) The shutter is in particular formed from an electrically insulating material such as plastic. For example, at least one portion of the shutter 58 comprises a grooved area 63 helping to stiffen said shutter. The grooved area 63 extends in particular over the radially extending portion 61.

(40) FIG. 5 shows another example of a cover 42 not comprising a portion specifically forming a gripping means 55. The other features and elements forming this cover are identical or essentially identical to those disclosed in the description of the first example of a cover described with reference to FIGS. 2 and 3.

(41) Thus, in this second example of a cover 42, the open region 44 extends over at least 95% of the axial end area of the inner portion of the cover 42. In other words, in this instance, the entire area of the inner portion is open, apart from the areas forming the axial protection portions 47 and the areas forming the fastening portions 53. In this example, the open region 44 is formed from a single axial opening 45.

(42) In one variant of this second exemplary embodiment, which is not shown, it is conceivable to modify the fastening portions 53 and the axial protection portions 47 so that they do not extend into the inner portion of the cover and to therefore further increase the opening percentage of the open region in order to obtain an opening percentage equal to 100%.

(43) The present invention is applicable in particular in the field of alternators, starter-alternators or reversible machines, but it could also apply to any type of rotating machine.

(44) Naturally, the preceding description has been given only by way of example and does not limit the field of the present invention, from which a departure would not be made by replacing the various elements with any other equivalents.

Claims

1. A cover for a rotating electric machine comprising an axis of rotation, the cover being intended to be arranged on an inverter of said machine and comprising: a skirt extending in an axial direction and forming a perimeter of the cover, the perimeter defining an inner portion of the cover; and an open region arranged in the inner portion and comprising at least one axial opening, wherein the open region extends over at least 50% of the inner portion of the cover, and wherein the skirt forms a closed perimeter and comprises a plurality of radial protection portions, each radial protection portion being intended to be arranged radially opposite a phase connection of the machine, a plurality of connecting portions extending circumferentially between two radial protection portions of the plurality of radial protection portions, each connecting portion having a height in an axial direction that is less than a height in the axial direction of each of the radial protection portions, and a plurality of primary radial openings formed between two radial protection portions of the plurality of radial protection portions such that two radial protection portions of the plurality of radial protection portions are separated from each other by one of the plurality of connecting portions and one of the plurality of primary radial openings.

2. The cover as claimed in claim 1, wherein the open region extends over at least 70% of the inner portion of the cover.

3. The cover as claimed in claim 2, wherein the open region comprises several axial openings.

4. The cover as claimed in claim 2, further comprising an axial protection portion intended to be arranged axially opposite at least one phase connection of the machine, the axial protection portion extending into the inner portion.

5. The cover as claimed in claim 2, further comprising a fastening portion, the fastening portion being intended to cooperate with a fastening element in order to mount the cover on an inverter of

the rotating electric machine, the fastening portion extending into the inner portion.

6. The cover as claimed in claim 2, further comprising gripping means extending into the inner portion.

7. A rotating electric machine comprising an inverter and a cover as claimed in claim 2, the cover being mounted on the inverter.

8. The cover as claimed in claim 1, wherein the open region comprises several axial openings.

9. The cover as claimed in claim 1, further comprising an axial protection portion intended to be arranged axially opposite at least one phase connection of the machine, the axial protection portion extending into the inner portion.

10. The cover as claimed in claim 1, further comprising a fastening portion, the fastening portion being intended to cooperate with a fastening element in order to mount the cover on an inverter of the rotating electric machine, the fastening portion extending into the inner portion.

11. The cover as claimed in claim 1, further comprising gripping means extending into the inner portion.

12. A rotating electric machine comprising an inverter and a cover as claimed in claim 1, the cover being mounted on the inverter.

13. The machine as claimed in claim 12, wherein the inverter comprises a capacitor bank and in that the open region is arranged so as to be at least partially axially aligned with the capacitor bank.

14. The machine as claimed in claim 13, wherein the inverter comprises at least one heat sink plate, said plate forming, at least partially, one axial end of said inverter and being arranged at least partially facing the open region.

15. The machine as claimed in claim 12, wherein the inverter comprises a heat sink plate forming, at least partially, one axial end of said inverter and being arranged at least partially facing the open region.

16. The machine as claimed in claim 12, further comprising a shutter arranged at one axial end of the inverter, the shutter comprising a radially extending portion arranged opposite a capacitor bank of the inverter.
